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Advancing smart aquaculture: Cost-efficient strategies for climbing perch cultivation using AI-based models

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ABSTRACT

This study introduces a hybrid AI-based optimization framework to enhance climbing perch aquaculture in smart farming systems, targeting improvements in both productivity and cost-efficiency. By integrating Taguchi experimental design with reinforcement learning and metaheuristic algorithms, the model identifies critical environmental and operational parameters that influence fish growth and economic outcomes. A multi-objective regression approach is applied to optimize key factors including water temperature, dissolved oxygen, pH, feeding schedules, and stocking densities. The RL-AMIS method, which combines Pareto front analysis and TOPSIS, successfully balances trade-offs between cost and productivity. Experimental results show a 15.3 % reduction in operational costs and a 17.8 % improvement in growth efficiency compared to conventional methods. These findings demonstrate that AI-driven optimization can enhance scalability and adaptability in aquaculture systems, supporting sustainable production strategies.

1. Introduction

Climbing perch (Anabas spp.) plays a vital role in both the economy and ecology of Southeast Asia, serving as a significant source of income for aquaculture communities and contributing to freshwater ecosystem balance. Ecologically, it helps control insect and invertebrate populations, thereby supporting biodiversity and water quality [1,2]. Rising consumer demand, coupled with environmental pressures, necessitates sustainable aquaculture practices to ensure long-term species availability and reduced ecological disruption.

Recent developments in closed-system smart farming aim to enhance productivity and lower costs by controlling key variables such as water quality, temperature, and oxygen levels (Satheesh et al., 2023; [3]). Smart technologies that enable real-time environmental monitoring represent a significant advancement in achieving consistent, cost-effective production while minimizing resource waste.

Although studies on biofloc technology and selective breeding have contributed to climbing perch aquaculture [4,5], previous research has yet to comprehensively integrate complex environmental variables—such as multi-level water quality parameters, light exposure, and feeding frequency—with genetic diversity in a unified optimization framework.

To address the existing research gap, our study integrates Taguchi experimental design with reinforcement learning and metaheuristic algorithms within an Artificial Multiple Intelligence System (AMIS), forming a comprehensive AI-based optimization framework (RL-AMIS). This novel approach captures the complex interdependencies among environmental and operational parameters in smart aquaculture systems, particularly for climbing perch cultivation. By systematically refining a wide range of growth factors, the framework aims to strike an optimal balance between maximizing biological productivity and minimizing operational costs—an essential equilibrium for the economic and sustainable advancement of aquaculture in Southeast Asia (Satheesh. et al., 2023; [1–3]).

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The structure of this research is laid out as follows: Section 2 will delve into the literature review, providing a foundation for the study. This is followed by Sections 3 and 4, where we will detail the research methodology and present the results, respectively. The analysis then culminates in Sections 5 and 6 with a comprehensive discussion of the findings and the conclusion, shedding light on the implications and potential future directions of our work.

2. Literature review

2.1. Climbing perch in Southeast Asia: economic cornerstone and ecological pillar amidst challenges in aquaculture sustainability

The economic and ecological importance of climbing perch in Southeast Asia is profound, serving as a cornerstone for both local economies and freshwater ecosystems. Climbing perch, a species adaptable to varying environmental conditions, is vital for local fisheries and aquaculture due to its high nutritional value and market demand [6]. The fish contributes significantly to food security, livelihoods, and the economic stability of rural communities, emphasizing its economic importance [7]. Ecologically, climbing perch plays a crucial role in maintaining the health and balance of freshwater ecosystems. Its presence indicates the water body's health and contributes to the biodiversity essential for ecosystem stability and resilience [8].

However, the aquaculture of climbing perch faces several challenges, especially with the increasing demand for this fish. Unsustainable fishing practices and environmental degradation pose significant threats to the sustainable aquaculture of climbing perch [9]. These challenges highlight the urgent need for innovative farming practices that can ensure the species' long-term viability [10,11]. Sustainable management strategies are crucial to address ecological, social, and economic issues arising from aquaculture activities, underscoring the necessity of adopting practices that minimize environmental impacts while maximizing production efficiency.

Jamu et al. [12] discusses the challenges in climbing perch aquaculture, including the need for sustainable management to address the ecological, social, and economic issues associated with unsustainable fishing practices. Faruk et al. [13] outlines factors crucial for the success of climbing perch aquaculture, such as suitable culturing location, water quality, seed quality, feeding management, disease control, and proper harvesting practices. These studies underscore the dual value of climbing perch to Southeast Asia's local economies and ecosystems, as well as the critical need for sustainable aquaculture practices to overcome the challenges faced in climbing perch farming.

For further reading on the economic and ecological importance of climbing perch and the challenges in its aquaculture, please refer to the works cited Chakraborty [9] and Tahir et al. [14]. These references provide a comprehensive overview of the significance of climbing perch and the imperative for sustainable aquaculture practices.

2.2. Advancements in smart farming and genetic enhancements revolutionize aquaculture: a focus on climbing perch cultivation

Recent innovations in smart aquaculture—particularly in IoT-enabled and AI-driven systems—have enabled better control over water temperature, oxygen levels, and feeding cycles in closed systems [15]. Technologies such as biofloc systems, which recycle waste into usable nutrients, and genetic enhancements have shown promise in improving growth rates in climbing perch [16,17]. However, most studies remain generalized, with little critical focus on species-specific applications.

Notably, while biofloc systems promote water efficiency and protein recycling, their integration with AI-based, real-time feedback systems for climbing perch remains largely unexplored. For instance, feed modulation and light control based on predictive analytics are underutilized. Although aquaponics and genomic selection have enhanced sustainability in other species [18,19], such techniques have not been customized to the behavioral and physiological traits of Anabas testudineus.

These advancements underscore the potential of smart farming technologies and genetic enhancements to address the challenges previously mentioned in climbing perch aquaculture. Through the implementation of these innovative practices, aquaculture can achieve greater sustainability, efficiency, and productivity, contributing positively to global food security and environmental conservation.

Recent advancements in artificial intelligence have significantly enhanced aquaculture monitoring and management. For instance, AI-driven systems are increasingly used for fish tracking, behavior analysis, and real-time environmental control [20,21]. These developments have made precision aquaculture more accessible and scalable in both research and commercial applications.

Moreover, sustainability assessments of closed-loop aquaculture systems are gaining attention, with region-specific indicators and environmental metrics being proposed to support long-term viability ([22]; WWF and Climate Focus, 2024). These perspectives align with the integrated design approach presented in this study.

2.3. Enhancing aquaculture through artificial intelligence and machine learning: applications, predictions, and future prospects

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as transformative technologies in aquaculture, offering innovative tools for optimizing environmental conditions, predicting growth performance, and improving decision-making processes. Their integration has led to smarter and more adaptive systems for sustainable fish farming.

One of the critical applications of AI and ML in aquaculture includes the use of machine learning for tasks such as measuring size, weight, grading, disease detection, and species classification [23]. Moreover, deep learning, a subset of machine learning, is emphasized for its capability to automatically extract features from raw data, showing promising results in detection, classification, counting, and prediction tasks [24].

Furthermore, employing AI and ML for environmental factor prediction models in aquaculture has shown to accurately predict water quality, thus enhancing water quality management and reducing aquaculture risks [25]. Additionally, IoT and Machine Learning technologies have been used to monitor and forecast water quality parameters, aiding in timely decision-making and potentially preventing production losses [26].

Advanced ML models such as KNN and decision tree regression have shown high predictive accuracy—up to 99 %—in estimating fish growth under controlled settings [27]. Similarly, Dhal and Kar [28] benchmarked various AI techniques for growth forecasting in aquaponics, emphasizing model performance in fluctuating environmental scenarios.

Despite this progress, real-time, closed-loop AI systems—especially those employing reinforcement learning (RL)—remain underutilized in aquaculture. Son and Jeong [29] recently introduced RL-based autonomous feeding mechanisms, highlighting significant potential to reduce feed waste and improve conversion ratios. However, no prior studies have applied such systems to climbing perch aquaculture.

Additionally, Hasan et al. [30] employed deep learning for predicting mortality and growth in small-scale sensor-driven fish farms, illustrating the scalability of AI solutions even in low-resource settings. Yet, these models often address singular objectives (e.g., growth or quality) rather than multi-objective trade-offs between biological performance and economic cost.

The broader implications of employing AI and ML in aquaculture, particularly through the use of Taguchi design, reinforcement learning, and metaheuristics, contribute significantly to the advancement of the field. These technologies fill existing gaps by offering solutions for optimizing aquaculture conditions, enhancing productivity, and

reducing operational costs, thereby supporting the sustainability of aquaculture practices.

2.4. The challenge of balancing economic viability and environmental sustainability: aquaculture's response to global fish protein demand

The aquaculture sector faces the urgent challenge of balancing productivity and cost-efficiency to ensure long-term viability in the context of rising global demand for fish protein. As reliance on aquaculture increases, economically sound and environmentally sustainable practices become paramount.

Researchers are refining aquaculture methods to minimize ecological impact. Sustainable approaches like polyculture and offshore facilities demonstrate potential for enhanced sustainability [31]. However, the effectiveness of existing certification standards in promoting ecosystem health remains a subject of debate, suggesting the need for management strategies that prioritize local community needs and tailor aquaculture zone management [32].

The nexus of aquaculture's economic significance and its environmental footprint is a central theme in the literature. Studies examine environmental consequences and sustainability issues, providing crucial insights for comprehensive impact assessments and policy decisions [33]. Cross-national policy analyses, such as those focusing on Singapore and Norway, illuminate how feed production and utilization significantly affect aquaculture's environmental performance [34].

Achieving sustainable seafood production requires a multifaceted approach. Challenges include the need for comprehensive understanding, innovative solutions, and enabling environments that support sustainable development. A balanced strategy must encompass environmental, social, and economic considerations [35]. Thailand exemplifies a context where specific environmental threats – habitat destruction, pollution, and wider ecological effects – demand robust management interventions to safeguard aquaculture's future [36].

To shape a sustainable and productive aquaculture sector, researchers propose four actionable goals: improved management practices, localized decision-making with emphasis on human capacity building, enhanced risk management, and systems that differentiate and promote sustainably produced seafood [37]. Additionally, federal environmental agencies, legislative reform, and stricter regulations are seen as key to mitigating negative impacts and fostering sustainable aquaculture [38].

The overarching objective of harmonizing economic viability with environmental sustainability is inextricably linked to aquaculture's ability to sustainably address the global need for fish protein. Integrating innovative practices with policies that address environmental, economic, and social dimensions will position the industry as a vital contributor to food security while safeguarding vulnerable ecosystems.

3. Research methodology

To improve clarity and reproducibility, this section is divided into two parts. Section 3.1 explains the experimental design and data collection procedures used in the climbing perch aquaculture study. Section 3.2 presents the optimization strategy using a hybrid Reinforcement Learning–Artificial Multiple Intelligence System (RL-AMIS) to determine cost-effective aquaculture parameters.

3.1. Experimental design and data collection

The experimental design framework utilizes the Taguchi method in aquaculture research, particularly for exploring complex growth factors related to climbing perch cultivation. Taguchi design is a robust statistical approach that maximizes parameter estimation efficiency by examining multiple variables concurrently while minimizing the number of experimental trials. Compared to a full factorial design which would require 19,683 trials (3^9 combinations) for nine parameters Taguchi significantly reduces the number of experiments to 27 runs, making it practical under real-world constraints such as time, cost, and resource availability. While response surface methodology (RSM) can also reduce runs, it is still less feasible at this scale and assumes fewer factors than addressed in this study.

Taguchi offers several key advantages: efficient exploration of the parameter space, flexibility in prioritizing significant factors, precision in identifying variations that affect fish growth, and the ability to operate within budget and time constraints. Although Taguchi design assumes interaction effects are negligible or less critical, this limitation is mitigated in our study through a subsequent multi-objective optimization process (RL-AMIS), which accounts for nonlinear interactions among variables. Thus, the integration of Taguchi with AI-based optimization provides a complementary framework for both experimental feasibility and computational robustness. The selected factors and their corresponding levels are shown in Table 1.

To ensure clarity and consistency throughout the methodology and optimization framework, this section defines all key symbols, parameters, and mathematical variables used in the regression models, the RL-AMIS optimization process, and multi-objective decision-making. These notations serve as the foundation for the experimental design, computational modeling, and performance evaluation. A summary of the notations and their corresponding definitions is presented in Table 2.

To investigate the effects of multiple environmental and management factors on climbing perch growth, we implemented 27 carefully structured treatments using Taguchi's orthogonal array. Each treatment varied key parameters—including species, water temperature, dissolved oxygen, pH, feeding frequency, water quality indicators, light exposure, stocking density, and growth period—to capture diverse aquaculture conditions. Table 3 summarizes these combinations, enabling a systematic analysis of both individual and interaction effects, and supporting the identification of optimal cultivation strategies.

In the conducted study, the growth performance of climbing perch, Anabas testudineus, within technologically advanced aquaculture systems-specifically, smart farming of dimensions 1 m × 1 m × 1m-was rigorously evaluated. The experiment aimed to elucidate the effects of nine critical environmental and operational parameters on the fishes' fresh weight gain, a primary indicator of growth efficiency. These parameters encompassed Climbing Perch Species, Water Temperature, Dissolved Oxygen Levels, pH Levels, Feed Type, Water Quality Parameters, Light Exposure, Stocking Density, and Growing Periods.

Moreover, the economic feasibility of each experimental setup was assessed by calculating the total operational cost. This cost comprised two major components: (1) the electricity consumption for regulating Water Temperature, Dissolved Oxygen Levels, Water Quality Parameters, and Light Exposure throughout the Growing Periods, and (2) other operational costs, including the procurement of fish, oxygen, chemical substances for pH level adjustment, feed, installation of lighting systems, and overall management of the growing periods.

Utilizing Design-Expert software, a comprehensive regression model was developed to quantify the relationship between the aforementioned parameters (independent variables) and two key response variables: (1) the fresh weight of the climbing perch, reflecting growth performance, and (2) the total operational cost, indicating economic efficiency. The regression model aimed to identify significant factors and their interactions, providing insights into optimizing both biological and economic outcomes.

Example of the data collected from the experiment is shown in Table 4 and example of regression model is shown in Table 5.

Based on the data presented in Table 4, the fresh weight of the fish was measured using a digital weighing machine. The operating cost was calculated by aggregating key expense components in aquaculture operations, including utilities (electricity and water), feed procurement and storage, labor wages, water quality management chemicals, lighting systems, and general overheads (e.g., administrative costs and equipment maintenance). These values were then used to construct a

Table 1
Experimental parameters and levels for climbing perch cultivation.

Parameters	Definitions	Types of Parameters	Levels of Parameters	Notes	References
Climbing Perch Species	Species of Climbing Perch predominant in the habitat.	Category	1. Anabas testudineus (Thai Climbing Perch) 2. Anabas oligolepis (Arched Climbing Perch) 3. Anabas senegalensis (Senegal Climbing Perch)	1. Most widely found and well-known, native to Thailand. 2. Common in central and northeastern Thailand, more arched body shape. 3. Introduced species, more common in other regions of Southeast Asia.	[39]
Water Temperature	The thermal conditions within the tank.	Category	Low: 24 °C, Medium: 27 °C, High: 30 °C	Optimal range affects growth and health. Adjust based on seasonal changes.	[5]
Dissolved Oxygen Levels	The amount of oxygen available in water.	Category	Low: 3 mg/L, Medium: 5 mg/L, High: 7 mg/L	Critical for metabolism and preventing stress.	[40]
pH Levels	Measure of the acidity or alkalinity of water.	Category	Low: 6.0, Medium: 7.0, High: 8.0	Affects fish health and nutrient availability.	[41]
Feeding Schedule	Types of feed given to fish and the frequency of feeding.	Category	Low: 2 times/day, Medium: 3 times/day, High: 4 times/day	Influences growth rates; more frequent feedings may promote faster growth.	[42]
Water Quality Parameters/ Ammonia	Concentrations of ammonia, nitrites, and nitrates.	Category	Low: 0.01 mg/L, Med: 0.1 mg/L, High: 0.2 mg/L	High levels are toxic; maintaining low to medium levels is crucial.	[43]
Water Quality Parameters/ Nitrites		Category	Low: 0.25 mg/L Med: 0.5 mg/L High: 1 mg/L		
Water Quality Parameters/ Nitrates		Category	Low: 10 mg/L Med: 20 mg/L, High: 40 mg/L		
Light Exposure	The amount and intensity of light the fish are exposed to.	Category	Low: 6 h/day, Medium: 9 h/day, High: 12 h/day	Affects physiological processes and growth. Adjust according to growth stages.	[44]
Stocking Density	The number of fish housed per unit volume of water.	Category	Low: 0.5 fish/L, Medium: 1 fish/L, High: 1.5 fish/L	Too high density can lead to stress and disease. Optimize for growth.	[45]
Growing Periods	The duration from stocking to harvest.	Category	Early: 5 months, Standard: 7 months, Extended: 9 months	Early harvest might result in smaller size but quicker turnover. Standard and extended periods allow for larger size but require more resources.	[46]

regression model to forecast both fresh weight and operating cost outcomes.

For modeling purposes, all regression coefficients (β) were estimated using Design-Expert software. The regression model incorporates 11 predictors representing aquaculture parameters, as shown in Equation (1). Notably, Climbing Perch Species (β_1) is a categorical variable with three levels—Anabas testudineus, Anabas oligolepis, and Anabas senegalensis. To include it in the linear regression model, we applied dummy variable encoding by introducing two binary variables: one for Anabas oligolepis and another for Anabas senegalensis, while Anabas testudineus was used as the reference group. This approach allows the model to estimate species-specific effects while maintaining interpretability and compatibility with linear regression analysis.

The final prediction equation is shown in Equation (1), with the corresponding regression coefficients provided in Table 5. Furthermore, both models demonstrated strong goodness-of-fit metrics. The fresh weight model achieved an R^2 of 0.921 with an RMSE of 3.52 kg, while the operating cost model yielded an R^2 of 0.906 and an RMSE of 114.47 Baht. These results validate the models' predictive capability and their reliability in supporting aquaculture decision-making.

$$\textit{Fresh Weight/Operating cost} = \beta 0 + \beta 1 \times \textit{Climbing Perch Species} + \beta 2 \times \textit{Water Temperature} + \beta 3 \times \textit{Dissolved Oxygen Levels} + \beta 4 \times \textit{pH Levels} + \beta 5 \times \textit{Feed Schedule} + \beta 6 \times \textit{Water Quality Ammonia} + \beta 7 \times \textit{Water Quality Nitrites} + \beta 8 \times \textit{Water Quality Nitrates} + \beta 9 \times \textit{Light Exposure} + \beta 10 \times \textit{Stocking Density} + \beta 11 \times \textit{Growing Periods} + \epsilon \quad (1)$$

The objective function models outlined herein will undergo optimization employing a hybrid approach that integrates reinforcement learning and the artificial multiple intelligent system (AMIS), commonly referred to as "RL-AMIS." The optimization process will be conducted within the bounds of Equation (2) to Equation (10) which is shown in Table 6.

The experiment will be executed in the smart fishery farming as shown in Fig. 1. illustrates a comprehensive smart farming system tailored for Climbing Perch, incorporating both hardware and software to foster ideal growth environments. The hardware consists of durable fish tanks and sensors that monitor water quality—temperature, pH, oxygen, and nutrient levels. Central to this setup is a control unit (such as a PLC, Arduino, or Raspberry Pi) that interprets sensor data to regulate the environment using actuators. These actuators manage water temperature, oxygenation, pH levels, and feeding schedules, complemented by adjustable LED lights and monitoring cameras for observing fish health.

On the software front, data acquisition software gathers sensor inputs, which control software then analyzes to operate the actuators efficiently. A database system archives this data, which is further examined using analytical tools like MATLAB, R, or Python to optimize growth conditions. Users can monitor and adjust settings in real-time via a user-friendly interface.

Setting up this smart farming system involves assembling hardware, installing software, and conducting thorough calibration to ensure operational efficiency. Ongoing data analysis facilitates environmental fine-tuning, while regular maintenance supports system longevity. This integrated approach enhances climbing perch production by balancing productivity, cost-effectiveness, and ethical aquaculture practices. The smart farming system used in this research is illustrated in Figure 1.

3.2. Optimization strategy

The RL-AMIS framework, as illustrated in Fig. 2, comprises five

Table 2
Notations and definitions of variables used throughout the study.

Symbol / Notation	Description
FW	Fresh weight of climbing perch (kg) – response variable
OC	Operational and material cost (Baht) – response variable
CPS	Climbing perch species: A. testudineus, A. oligolepis, A. senegalensis
WT	Water Temperature (°C)
DOL	Dissolved Oxygen Level (mg/L)
pH	pH level of the water
FS	Feeding Schedule (times/day)
WQA	Water Quality – Ammonia (mg/L)
WQNi	Water Quality – Nitrites (mg/L)
WQNa	Water Quality – Nitrates (mg/L)
LE	Light Exposure (hours/day)
SD	Stocking Density (fish per liter)
GP	Growing Period (months)
β_0 to β_{11}	Regression coefficients for linear prediction models
X_{ij}^t	Encoded value of the j-th parameter in the i-th work package at iteration t (used in RL-AMIS)
Y_{ij}^t	Decoded parameter value corresponding to X_{ij}^t
Z_n	Objective function value (n = 1 for productivity, n = 2 for cost)
f_i^{t*}	Aggregated objective value of solution i at iteration t
HV	Hypervolume indicator – measures convergence and diversity of Pareto front
ARP	Average Ratio of Pareto-optimal solutions – performance evaluation metric
IB	Intelligence Box – algorithmic module used in RL-AMIS
WP	Work Package – solution vector in optimization process
TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution – decision ranking method
FW	Fresh weight of climbing perch (kg) – response variable
OC	Operational and material cost (Baht) – response variable
CPS	Climbing perch species: A. testudineus, A. oligolepis, A. senegalensis
WT	Water Temperature (°C)
DOL	Dissolved Oxygen Level (mg/L)
pH	pH level of the water

integrated stages designed to optimize aquaculture parameters for climbing perch by combining reinforcement learning (RL) with multiple intelligent heuristics (referred to as Intelligence Boxes or IB). The process begins with the generation of candidate solutions known as work packages (WP), each encoding normalized values of nine critical input parameters (e.g., water temperature, pH level, feeding schedule). These WPs are then directed to specific IBs using a reinforcement learning-based selection strategy that evaluates historical performance to guide intelligent decision-making. Each IB represents a distinct optimization algorithm (e.g., Genetic Algorithm, Water Cycle Algorithm, Momentum Search), and refines its assigned WPs accordingly to enhance solution quality. The refined WPs are subsequently evaluated based on two primary objectives maximizing fresh weight and minimizing operational cost using Pareto Front Optimization (PFO) to identify non-dominated solutions, followed by TOPSIS to rank alternatives relative to an ideal point. This process iterates with heuristic updates until a termination condition is met (e.g., predefined number of iterations or convergence criteria). Through this hybrid mechanism, RL-AMIS harnesses both the adaptive learning capabilities of RL and the targeted efficiency of diverse heuristic methods, resulting in a flexible and robust approach to identifying optimal conditions for sustainable and cost-effective aquaculture.

Fig. 2 illustrates how we adapted the AMIS framework to handle the challenge of multi-objective decision-making. We incorporated reinforcement learning (RL) to guide the method selection process. A specially designed variable plays a key role in determining which solutions progress through the framework's iterative process which will be explained as follows.

3.2.1. *Developing the initial work package set (WPs)*

The initial phase involves generating a set of work packages (WPs). These WPs are created using real number encoding within the range of 0 to 1. Each WP represents a member of the AMIS population and has "D" dimensions. In our experiment, these dimensions correspond to ten beta values (ranging from [specify a range, e.g., 0.1 to 0.9]; β_0 to β_{10}). For example, a 10-dimensional WP might look like this: {0.81, 0.03, 0.41, 0.60, 0.09, 0.98, 0.49, 0.88, 0.38}. Let's denote this number by X_{ijt}, which indicates the value of WP 'i' at position 'j' in time 't'. These numerical values are decoded to determine the corresponding aquaculture parameters such as water temperature, pH level, and feeding frequency

Table 3
Experimental treatments for assessing growth parameters in climbing perch.

T	CPS (Species)	WT (°C)	DOL (Mg/L)	pH Levels	FS (Times/Day)	WQA (Mg/L)	WQNi (Mg/L)	WQNa (Mg/L)	LE (Hours/Day)	SD (Fish/L)	GP (Months)
1	Testudineus	24	3	6	2	0.01	0.25	10	6	0.5	5
2	Testudineus	24	3	6	3	0.10	0.50	20	9	1	7
3	Testudineus	24	3	6	4	0.20	1.00	40	12	1.5	9
4	Testudineus	27	5	7	2	0.01	0.25	20	9	1	9
5	Testudineus	27	5	7	3	0.10	0.50	40	12	1.5	5
6	Testudineus	27	5	7	4	0.20	1.00	10	6	0.5	7
7	Testudineus	30	7	8	2	0.01	0.25	40	12	1.5	7
8	Testudineus	30	7	8	3	0.10	0.50	10	6	0.5	9
9	Testudineus	30	7	8	4	0.20	1.00	20	9	1	5
10	Oligolepis	24	5	8	2	0.10	1.00	10	9	1.5	5
11	Oligolepis	24	5	8	3	0.20	0.25	20	12	0.5	7
12	Oligolepis	24	5	8	4	0.01	0.50	40	6	1	9
13	Oligolepis	27	7	6	2	0.10	1.00	20	12	0.5	9
14	Oligolepis	27	7	6	3	0.20	0.25	40	6	1	5
15	Oligolepis	27	7	6	4	0.01	0.50	10	9	1.5	7
16	Oligolepis	30	3	7	2	0.10	1.00	40	6	1	7
17	Oligolepis	30	3	7	3	0.20	0.25	10	9	1.5	9
18	Oligolepis	30	3	7	4	0.01	0.50	20	12	0.5	5
19	Senegal	24	7	7	2	0.20	0.50	10	12	1	5
20	Senegal	24	7	7	3	0.01	1.00	20	6	1.5	7
21	Senegal	24	7	7	4	0.10	0.25	40	9	0.5	9
22	Senegal	27	3	8	2	0.20	0.50	20	6	1.5	9
23	Senegal	27	3	8	3	0.01	1.00	40	9	0.5	5
24	Senegal	27	3	8	4	0.10	0.25	10	12	1	7
25	Senegal	30	5	6	2	0.20	0.50	40	9	0.5	7
26	Senegal	30	5	6	3	0.01	1.00	10	12	1	9
27	Senegal	30	5	6	4	0.10	0.25	20	6	1.5	5

Table 4
Optimization of aquaculture parameters for anabas species using Taguchi experimental design.

T	CPS (Species)	WT (°C)	DOL (Mg/L)	pH Levels	FS (Times/Day)	WQA (Mg/L)	WQNi (Mg/L)	WQNa (Mg/L)	LE (Hours/Day)	SD (Fish/L)	GP (Months)	Fresh Weight (Kg)	Operating Cost (Baht)
1	Testudineus	24	3	6	2	0.01	0.25	10	6	0.5	5	52.13	2595.92
2	Testudineus	24	3	6	3	0.10	0.50	20	9	1	7	50.50	3281.16
3	Testudineus	24	3	6	4	0.20	1.00	40	12	1.5	9	42.25	4057.42
4	Testudineus	27	5	7	2	0.01	0.25	20	9	1	9	47.53	3379.95
5	Testudineus	27	5	7	3	0.10	0.50	40	12	1.5	5	37.05	3443.22
6	Testudineus	27	5	7	4	0.20	1.00	10	6	0.5	7	56.55	2955.93
7	Testudineus	30	7	8	2	0.01	0.25	40	12	1.5	7	34.08	3542.01
8	Testudineus	30	7	8	3	0.10	0.50	10	6	0.5	9	52.45	3037.33
9	Testudineus	30	7	8	4	0.20	1.00	20	9	1	5	51.85	3103.56
10	Oligolepis	24	5	8	2	0.10	1.00	10	9	1.5	5	57.30	3044.73
11	Oligolepis	24	5	8	3	0.20	0.25	20	12	0.5	7	44.55	3246.01
12	Oligolepis	24	5	8	4	0.01	0.50	40	6	1	9	24.28	3044.36
13	Oligolepis	27	7	6	2	0.10	1.00	20	12	0.5	9	65.45	4095.16
14	Oligolepis	27	7	6	3	0.20	0.25	40	6	1	5	32.60	2853.07
15	Oligolepis	27	7	6	4	0.01	0.50	10	9	1.5	7	70.83	3652.27
16	Oligolepis	30	3	7	2	0.10	1.00	40	6	1	7	25.75	2880.82
17	Oligolepis	30	3	7	3	0.20	0.25	10	9	1.5	9	52.50	3335.18
18	Oligolepis	30	3	7	4	0.01	0.50	20	12	0.5	5	51.13	3244.90
19	Senegal	24	7	7	2	0.20	0.50	10	12	1	5	64.35	3388.83
20	Senegal	24	7	7	3	0.01	1.00	20	6	1.5	7	55.33	3267.10
21	Senegal	24	7	7	4	0.10	0.25	40	9	0.5	9	33.45	3456.17
22	Senegal	27	3	8	2	0.20	0.50	20	6	1.5	9	35.00	2885.26
23	Senegal	27	3	8	3	0.01	1.00	40	9	0.5	5	26.13	2829.76
24	Senegal	27	3	8	4	0.10	0.25	10	12	1	7	53.00	3287.08
25	Senegal	30	5	6	2	0.20	0.50	40	9	0.5	7	33.55	3269.32
26	Senegal	30	5	6	3	0.01	1.00	10	12	1	9	72.78	4074.07
27	Senegal	30	5	6	4	0.10	0.25	20	6	1.5	5	49.30	2850.11

Table 5
Regression coefficients for predicting fresh weight and operating/material cost.

	β_0	β_1	β_2	β_3	β_4	β_5	β_6	β_7	β_8	β_9	β_{10}	β_{11}	R^2	RMSE
Fresh Weight (Kg)	71.287	- 0.148	-0.014	2.000	-5.042	0.972	12.292	7.976	-0.899	1.505	2.028	0.107	0.921	3.52
Operating and Material Cost (Baht)	52.460	- 0.083	-0.027	1.493	-4.082	0.851	-8.520	6.945	-0.009	3.006	4.034	3.009	0.906	114.47

Table 6
Parameter constraints for climbing perch species aquaculture experiment.

Parameters	Lower and upper-level constraint	Equation
Climbing Perch Species	Anabas testudineus/Anabas oligolepis/ Anabas senegalensis	(2)
Water Temperature	≥24 °C, and ≤30 °C	(3)
Dissolved Oxygen Levels	≥3 mg/L and ≤7 mg/L	(4)
pH Levels	≥6.0 and ≤8.0	(5)
Feed Type and Feeding Schedule	≥2 times/day and ≤4 times/day	(6)
Water Quality Parameters	Ammonia (≥0.01 mg/L, and ≤0.2 mg/L), Nitrites (≥0.25 mg/L and ≤1 mg/L), Nitrates (≥ 10 mg/L and ≤40 mg/L)	(7)
Light Exposure	≥6 h/day, and ≤12 h/day	(8)
Stocking Density	≥0.5 fish/L, and ≤1.5 fish/L	(9)
Growing Periods	≥5 and ≤9 months	(10)

based on the defined range for each factor. This decoding process enables the RL-AMIS framework to explore diverse environmental scenarios and guide the optimization of fish farming conditions.

The methodology that we will decode the coded value in 'WP' is called decoding methods. In the decoding method from the values in position of each WP can be executed, using the covert of value from minimum to maximum number allowed the varied from the setting of range of parameters shown in Table 1.

In our model, let Y_{ijt} represent the values of key factors that include Climbing Perch Species (CPS), Water Temperature (WT), Dissolved Oxygen Levels (DOL), pH Levels (pH), Feeding Schedule (FS), Water Quality Parameters (WQP), Light Exposure (LE), Stocking Density (SD), and Growing Periods (GP). Here, Y_{ijt} is the value attributed to factor 'j'

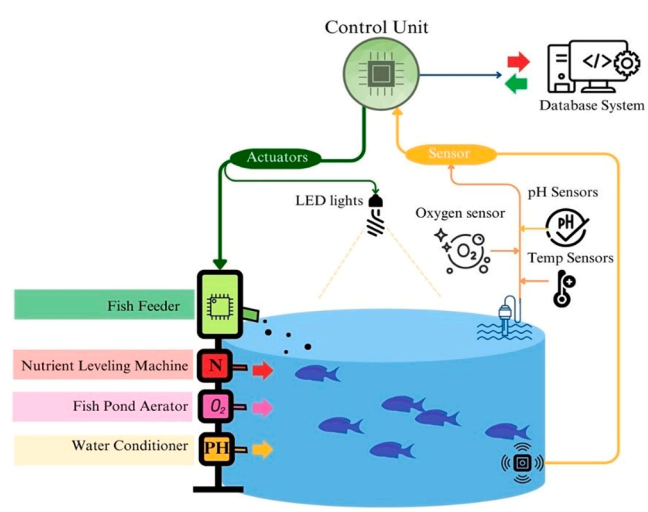

Fig. 1. Integrated aquaculture control center for sustainable climbing perch farming.

under Work Package 'i' during iteration 't' The calculation of Y_{ijt} for continuous variables is facilitated by Eq. (11), as detailed further down. The range of possible values for factor 'j' extends from S_j^{min} to S_j^{max}, indicating the minimum and maximum limits respectively, as specified in Table 6.

$$Y_{ijt} = S_j^{min} + \left[X_{ijt} \times \left(S_j^{max} - S_j^{min}\right)\right] \tag{11}$$

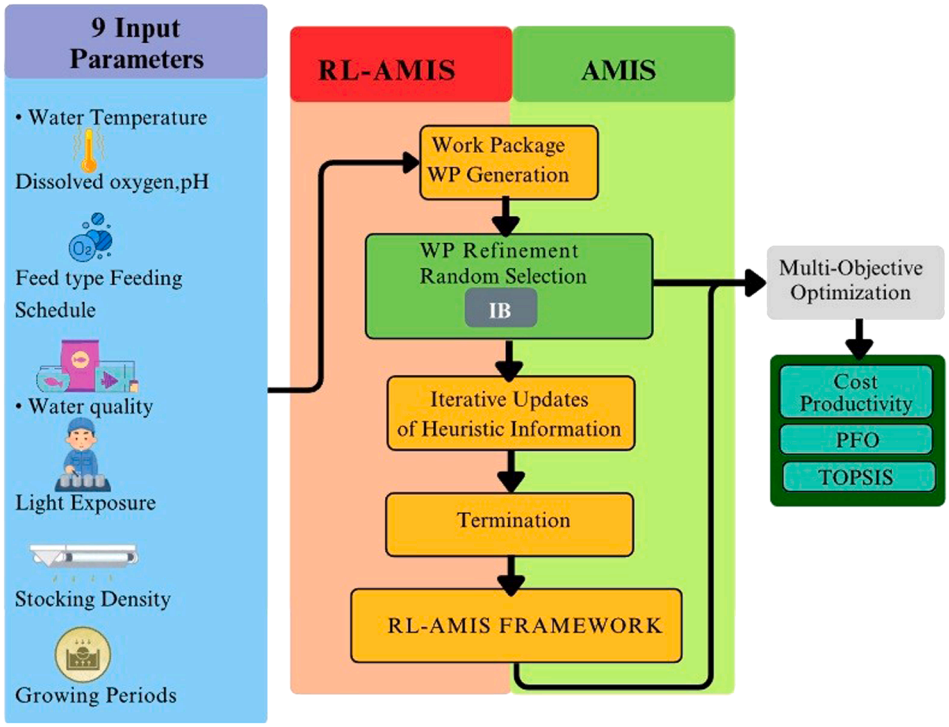

Fig. 2. Comprehensive view of the RL-AMIS framework.

For categorical variables, identification is governed by Eq. (12), where we define the total number of categories for factor 'j' by "M_j". The probability of selecting a specific category 'm', represented by p_{jm} , where 'm' is designated as the category number. The cumulative probability of choosing category 'm', enoted as $'C_{jm}'$,is calculated using Eq. (13). For a given work package 'i', the normalized value X_{ijt} the category selection for factor 'j'. if $X_{ijt} \in \left[C_{j(m-1)}\ , C_{jm}\right)$ then level 'm' is assigned to that factor. In cases where a categorical variable has three levels, the intermediate value is represented by 'S_j^{mid}'. All categorical levels and their corresponding values are detailed in Table 1.

$$p_{jm} = \frac{1}{M_j}\ \forall\, j = 1 \ldots J \text{ and } m = 1 \ldots M \tag{12}$$

$$C_{jm} = \sum_{l=1}^{m} p_{jl}\ \forall\, j = 1 \ldots J \text{ and } m = 1 \ldots M \tag{13}$$

The outcomes of the study, specifically Fresh Weight (FW in Kg) and Operation Cost (OC in), are calculated using Eqs. (14) and (15). We introduce ϑ^1 and ϑ^2 as the weights for our objectives (FW and OC), ensuring that their sum equals one ($\vartheta^1 + \vartheta^2 = 1$). Random factors φ^1, *and* φ^2 as, which vary between zero and one, influence the calculation. The indicators 1 and 2 represent different objectives and are denoted by 'n'. Eqs. (14) and (15) are utilized to compute the values of ϑ^1, and ϑ^2, and to define the objective function for a specific Work Package 'i' during iteration 't', represented as Z_{it}^n, is defined as $f\left(Z_{it}^n\right)$, the optimal function value f_{it}^* is determined by Eq. (15). It's important to note that φ^1, *and* φ^2 are randomly generated for each 'i' and 't', with 'k' acting as a placeholder for 'n' in these calculations.

$$\vartheta^n = \frac{\varphi^n}{\sum_{k=1}^{5} \varphi^k} \tag{14}$$

$$f_{it}^* = \sum_{n=1}^{2} \vartheta^n Z^n \tag{15}$$

3.2.2. Enhancing existing work package solutions through selected intelligence boxes

In this section, we discuss how the Adaptive Multi-Sensor Integration System (AMIS) framework is augmented through the incorporation of reinforcement learning (RL), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), and Pareto Front Optimization (PFO), leading to the development of RL-AMIS. In the RL-AMIS framework, each work package (WP) is treated as an individual agent. A predetermined number of WPs, denoted as NP, forms a group of agents referred to as an agent population (WP q, with 'q' representing numbers from 1 to NP). The term 'B' is used to indicate the number of intelligence boxes (IBs) integrated into the RL framework. Each cycle of reinforcement learning is identified as a stage, labeled as Stage 't', which reflects the iteration number 't'. Within this framework, the process for selecting WPs from the intelligence boxes (IB) environment is guided by a specific decision-making policy, which is defined by Eq. (16) in the context of the RL framework.

$$P_{bt} = \frac{F1N_{bt-1} + F2A_{bt-1} + F3I_{bt-1} + F4G_{bt-1}}{\sum_{b=1}^{B} F1N_{bt-1} + F2A_{bt-1} + F3I_{bt-1} + F4G_{bt-1}} \tag{16}$$

The selection probability of an Intelligence Box (IB) 'b' during iteration 't', represented as P_{bt}, is influenced by several key factors. These factors include the number of Work Packages (WPs) that chose IB 'b' in the previous iteration (N_{bt-1}) the average objective function value derived from WPs selecting IB 'b' in the prior iteration (A_{bt-1}), and a reward variable I_{bt-1} determined by Eq. (17). The selection of IB follows a probability function specified in Eq. (11) and is implemented utilizing the roulette wheel selection method. Constants 'F1′ and 'F2′ are assigned a value of 1, whereas 'F3′ and 'F4′ are both set to 2.

$$I_{bt} = I_{bt-1} + \begin{cases} 1 \text{ if IB } b \text{ holds the global best solution during iteration} t. \\ 0 \text{ otherwis} \end{cases}$$

$$\forall\, b = \{1, 2, \ldots, M), t = \{1, 2, \ldots, T\} \tag{17}$$

The variable G_{bt} represents the accumulated reward acquired through the utilization of IB 'b' while refining solutions, as determined

by the computational procedure outlined in Eq. (18). The parameter 'γ' spans a range of values from 0 to 1, with a specific value of 0.2 chosen for this experimental scenario. Additionally, ω_{bqt}, a binary variable, assumes a value of 1 when WP 'q' opt to employ IB 'b' during the 't-th' iteration and is assigned a value of 0 otherweise, as explicitly defined in Eq. (19).

$$G_{bt} = \frac{\sum_{q=1}^{NP}\omega_{bqt-1}G_{bt-1}}{\sum_{q=1}^{NP}\omega_{bqt-1}} + \frac{\sum_{q=1}^{NP}\omega_{bqt-2}\gamma R G_{bt-2}}{\sum_{q=1}^{NP}\omega_{bqt-2}} + \frac{\sum_{q=1}^{NP}\omega_{bqt-3}\gamma^2 G_{bt-3}}{\sum_{q=1}^{NP}\omega_{bqt-3}} + \ldots + \frac{\sum_{q=1}^{NP}\omega_{bq1}\gamma^{t-1}G_{b1}}{\sum_{q=1}^{NP}\omega_{bq1}} \ \forall\, b = \left\{1, 2, \ldots, M\right\}, t = \{1, 2, \ldots, T\} \tag{18}$$

$$\omega_{bqt}\begin{cases} 1 \ if \ WP \ q \ select \ IB \ b \ in \ iteration \ t \\ 0 \ otherwise \end{cases} \tag{19}$$

3.2.3. Improving the solution using selected IB

In our study, we apply a framework consisting of ten "intelligence boxes," a concept borrowed from prior research, specifically outlined in Eqs. (20) to (27). This framework allows us to track the progress of Work Packages (WPs), designated by 'p', across iterations, denoted by 'q'. Within this setup, the value of a WP at position 'l' during iteration 'q' is represented by U_{ijt}. We identify WPs using labels r_1,and r_2. Among all solutions we've found so far, U_j^{best} represents the one that delivers the best outcome. Additionally, we incorporate a random element in our calculations: α is a random variable that can take any value between 0 and 1, and $\mathbb{R}_{ijt}$ is a specific random number, also between 0 and 1, associated with WP 'p' at position 'l'. The equations mentioned earlier, 20 through 27, integrate techniques from a variety of optimization algorithms. These include the Cultural Algorithm (CA) as seen in Eqs. (20) and (21), described by Kechid and Drias [47], the Water Cycle Algorithm (WCA) in Eq. (22) from Meng et al. [48], the Momentum Search Algorithm (MSA) covered in Eqs. (23) and (24) by Shekhawat et al. [49], the Equilibrium Optimizer (EO) in Eq. (25) by Korashy et al. [50], the Genetic Algorithm (GA) in Eq. (26) from Cauvery and Viswanatha [51], and the Restart Algorithm in Eq. (27). We will compare the effectiveness of these algorithms with others in the experimental results section of our study.

$$U_{ijt} = U_{r_1jt-1} + \mathbb{R}_{\boldsymbol{ijt}} \tag{20}$$

$$U_{ijt} = 0.5 \times U_{r_1jt-1} + 0.5 \times U_{r_2jt-1} \tag{21}$$

$$U_{ijt} = U_{ijt-1} + \mathbb{R}_{\boldsymbol{ijt}} \times U_{r_1jt-1} \tag{22}$$

$$U_{ijt} = \alpha \times U_{ijt-2} + (1-\boldsymbol{\alpha}) \times U_{ijt-1} \tag{23}$$

$$U_{ijt} = U_{ijt-1} + \mathbb{R}_{\boldsymbol{ijt}} \times \left[\alpha \times U_{ijt-2} + (1-\boldsymbol{\alpha}) \times U_{ijt-1}\right] \tag{24}$$

$$U_{ijt} = U_{ijt-1} \times \left(\left(U_{r_1jt-1} - U_{ijt-1}\right) + \mathbb{R}_{\boldsymbol{ijt}}\right) + F \times \left(\frac{t}{T}\right) \tag{25}$$

$$U_{ijt} = \begin{cases} U_{pjt-1} \ if \ \mathbb{R}_{ijt} \le CR \\ U_j^{best} \ otherwise \end{cases} \tag{26}$$

$$U_{ijt} \ = \ \mathbb{R}_{\boldsymbol{ijt}} \tag{27}$$

In our methodology, we introduce a system where $\mathbb{R}_{\boldsymbol{ijt}}$ signifies random values assigned to Work Package 'i' at position 'j' during iteration 't'. Within this system, α is a predetermined factor specific to the Momentum Search Algorithm (MSA), set at 0.2. We denote 't' as the current iteration, with 'T' representing the maximum number of iterations allowed, and 'F' is the time factor scaling factor, preset at 5. Additionally, the crossover rate, CR, is established at 0.7. This approach enables us to adopt distinct methods for improvement in each iteration, significantly boosting the system's adaptability and robustness. Through the use of Eqs. (14) and (15), we calculate f_{it}^*, which helps us identify f_t^{best}. The latter symbolizes the best solution derived from all the f_{it}^* values, which can be precisely determined with Eq. (28). Our strategy's emphasis on flexibility and robustness is designed to facilitate the optimal selection of improvement methods at every step of the process.

$$f_t^{best} = \begin{cases} f_{it}^* \ if \ f_{it}^* \ge f_{t-1}^{best} \\ f_{t-1}^{best} \ otherwise \end{cases} \tag{28}$$

After updating the value of f_t^{best}, we see a corresponding change in U_j^{best}. This change aligns U_j^{best} with the value at U_{ijt}, which represents the best solution we've found. The process for updating U_{ijt} depends on meeting the criteria set forth in Eqs. (29) to (32). To make this clearer, we introduce the concept of ω_{ij}^n, a binary variable. This variable is set to one if we observe an improvement for objective 'i' in the current iteration 'q' over the previous iteration 'q-1'. If there's no improvement, ω_{ij}^n is set to zero. The term ω_{ij}^{total} then represents the total sum of ω_{ij}^n across all objectives 'n', giving us a cumulative measure of improvement.

$$\omega_{it}^1 = \begin{cases} 1 \ if \ f^1\left(U_{ijt}\right) \ge \ f^1\left(U_{ijt-1}\right) \\ 0 \ otherwiese \end{cases} \tag{29}$$

$$\omega_{it}^2 = \begin{cases} 1 \ if \ f^2\left(U_{ijt}\right) \ge \ f^2\left(U_{ijt-1}\right) \\ 0 \ otherwiese \end{cases} \tag{30}$$

$$\omega_{ij}^{total} = \sum_{n=1}^{2}\omega_{ij}^n \ \forall p, q \tag{31}$$

$$U_{ijt+1} = \begin{cases} U_{ijt} \ when \ \omega_{it}^{total} \ge 1 \\ U_{ijt-1} \ otherwise \end{cases} \tag{32}$$

3.2.4. Acquisition of pareto front and execution of TOPSIS analysis

In our analysis, the acquisition of the Pareto front is crucial for identifying and keeping the best, non-dominated solutions. This process involves examining the objectives for Work Package r, which are expressed as $f^1(U_r)$, and $f^2(U_r)$. We define the set of all possible solutions as ℜ, where each solution, u, is represented by a decision vector $u = (u_1, u_2, \ldots, u_i)$. The objective vectors corresponding to these decision vectors are denoted as f^v(u) = (f^1(u), f^2(u), …, f^V(u)). A solution u is considered to dominate another solution u' if, for every objective v from 1 to V, f^v(u) ≤f^v(u'). The process, named AMIS, continues iterating until it meets certain predefined stopping criteria and successfully delineates the Pareto front. Following this, we apply the TOPSIS method, which stands for "technique for order preference by similarity to the ideal solution," to pinpoint the most optimal set of parameters.

3.2.5. Iterative heuristic information update and recurrence of steps 4.2-4.5

In the heart of our algorithmic approach is the regular update of parameters critical to efficiency: N_{bt}, A_{bt}, I_{bt}, U_l^{best}, *and* $\mathbb{R}_{pl}$. This process of refinement unfolds in every cycle of the algorithm, methodically progressing through steps 1 to 4 until it reaches predefined stopping points. From our previous work, we noted that the algorithm completes its function after 3,000 cycles. This detail, highlighted in the RL-AMIS pseudocode, emphasizes the importance of this calibration and iterative improvement process within heuristic-based computational algorithms in modern research efforts.

3.2.6. Evaluation metrics for performance assessment

The Hypervolume (HV) metric is essential for assessing how closely solutions, or reference points, approach and spread around the Pareto front, as outlined in Eq. (33). HV values shed light on how well solutions match the ideal point and their distribution. Specifically, $HV\left(f^{ref}, X\right)$

measures the size of a set of solutions by comparing them to a reference objective, denoted as f^{ref}. for any given point X. This comparison uses the Lebesgue measure Λ(.) to evaluate the objectives from f_1 through f_n. Additionally, the Average Ratio of Pareto Optimal Solution (ARP) is a key measure for evaluating performance, calculated as per Eq. (34). ARP represents the ratio of Pareto optimal points to the total number of solutions. A higher ARP value suggests a denser cluster of Pareto points, indicating better performance of the algorithm. The ARP takes into account iterations $N_1, N_2, \ldots, N_k$ from Experiments 1 to k. Here, n_1, n_2, …, n_k represent the counts of Pareto optimal solutions in each iteration, and K is the total number of experiments, facilitating a thorough evaluation.

$$HV\left(f^{ref}, X\right) = \Lambda\left(\bigcup_{X_n \in X}\left[f_1(X_n), f_1^{ref}\right] \times \ldots \times \left[f_m(X_n), f_m^{ref}\right]\right), \tag{33}$$

$$ARP = \frac{\frac{n_1}{N_1} + \frac{n_1}{N_2} + \ldots + \frac{n_k}{N_k}}{K} \tag{34}$$

3.2.7. Comparative methods

In our research, we will evaluate the performance of RL-AMIS in comparison with several notable methodologies. These include the original AMIS framework developed by Pitakaso et al. [52], and other established methods like the Cultural Algorithm (CA) detailed by Kechid and Drias [47], the Water Cycle.

Algorithm (WCA) introduced by Meng et al. [48], the Momentum Search Algorithm (MSA) by Shekhawat et al. [49], the Equilibrium Optimizer (EO) from Korashy et al. [50], and the Genetic Algorithm (GA) as explained by Cauvery and Viswanatha [51]. Additionally, we plan to compare our experimental findings against the best results obtained through an approximation technique available in Design Expert software, which we refer to as "Best Experimental Result" (BER) and "Best Solution Obtained from Design Expert Software" (BSDE). To ensure a fair evaluation across all methods, each algorithm is allocated an equal computational budget of 60 min.

The Artificial Multiple Intelligence System (AMIS) is a metaheuristic framework that manages a population of candidate solutions called work packages (WPs). Each WP is iteratively refined using one metaheuristic, known as an Intelligence Box (IB), selected from a predefined pool. IB selection is probabilistic and based on a performance-weighted function combining usage count, average fitness, immediate reward, and accumulated reward, as defined in Eq. (16).

In RL-AMIS, each Intelligence Box (IB) receives a binary immediate reward Ibt Eq. (17) if it produces the global best solution in a given iteration. This reward contributes to a cumulative reward score Gbt Eq. (18) calculated using a discounted sum of past rewards with decay factor γ. The accumulated reward guides future IB selection, enabling the RL agent to favor consistently effective strategies.

Table 7 presents a consolidated comparison of the algorithmic configurations used across all optimization methods evaluated in this study. All algorithms were executed under consistent conditions, with a population size of 50 and a runtime of 60 min, ensuring fair performance comparison. For RL-AMIS and AMIS, a shared pool of Intelligence Boxes (IBs) including multiple variants of Cultural Algorithm (CA), Momentum Search Algorithm (MSA), Water Cycle Algorithm (WCA), Equilibrium Optimizer (EO), Genetic Algorithm (GA), and Restart—was employed. The table outlines the key control parameters for each algorithm, such as learning rates, mutation and crossover rates, selection strategies, and specific hyperparameters unique to each metaheuristic. This unified parameterization supports a transparent and reproducible benchmarking environment for climbing perch aquaculture optimization.

RL-AMIS, is a hybrid optimization framework specifically designed for multi-objective decision-making in complex environments. It operates by dynamically selecting from a set of metaheuristics referred to as Intelligence Boxes (IB) through reinforcement learning mechanisms. These IBs iteratively refine candidate solutions, known as Work Packages (WP), to explore the trade-off space between productivity (e.g., fish growth) and cost-efficiency. The performance of candidate solutions is assessed using Pareto Front Optimization and TOPSIS ranking techniques. The overall structure of this optimization process is summarized in Algorithm 1.

Table 7
Algorithmic parameters for each algorithm.

Algorithm	Parameter	Value / Description
All-Algorithms	Running Times	60 min
	Population Size	50
RL-AMIS	Selection Method	RL-guided Roulette Wheel
	Learning Rate (α)	0.01
	Discount Factor (γ)	0.2
	Exploration Rate (ε)	0.1
	Intelligence Boxes (IBs)	CA1-CA2, WCA, MSA-1, MSA-2, EO, GA, Restart
AMIS	Selection Method	Roulette Wheel with performance metrics
	Intelligence Boxes (IBs)	CA1-CA2, WCA, MSA-1, MSA-2, EO, GA, Restart
MSA	Strategy	Momentum-based search
	Weight Factor (α)	0.2
	Random Factor	[0,1]
WCA	Strategy	Flow-based optimization
	Number of Rivers	2–5
	Rain Rate (RR)	Probability ∈ [0.01 – 0.1]
EO	Strategy	Equilibrium-based dynamic search
	Time Scaling Factor (F)	5
	Randomization Factor (λ)	∈ [0.5, 1.0]
	Equilibrium Pool Size	4
GA	Crossover Rate	0.7
	Mutation Rate	0.05
	Selection Method	Elitism / Best preservation
CA	Influence Function	Adjust mutation/crossover ranges
	Acceptance Function	top 30 % of solutions based on fitness Z*
	Mutation Rate	0.7
	Crossover Rate	0.05

Algorithm 1: RL-AMIS: Reinforcement learning-Adaptive metaheuristic-based improvement system input.

```
T: Maximum number of iterations for the main algorithm loop
NP: Number of Work Packages (i.e., solutions in the population)
IB [1...B]: Intelligence Boxes of operators (e.g., {CA, WCA, MSA, EO, GA, Restart, ...})
Objectives: Fresh Weight (maximize) and Operating Cost (minimize)
γ (gamma) ← 0.2: Discount factor for Reinforcement Learning reward calculation (Eq. (18))
CR ← 0.7: Crossover Rate for Genetic Algorithm (GA) operator (Eq. (26))
α (alpha) ← 0.2: Weight factor for Magnetic Spring Algorithm (MSA) operator (Eq. (23))
F ← 5: Time scaling factor for Equilibrium Optimizer (EO) operator (Eq. (25))
Begin:
// — STAGE 1: Initial WP Generation — (Eqs. (11)–(15))
for i = 1 to NP: // Initialize & Evaluate each Work Package (WP_i)
X_i ← GenerateRandomNormalizedParams(D); Y_i ← DecodeParams (X_i, ParamRanges, Eqs. (11)-(13)
FW_i, OC_i ← EvaluateObjectives (Y_i, f1, f2) // Evaluate objectives
Z_it* ← CalculateFitness (FW_i, OC_i, ϖ1, ϖ2, Eqs. (14)-(15)) // Compute fitness
end for
// — STAGE 2: Main Iteration Loop —
for t = 1 to T: // For each iteration
for each WP_i in Population: // For each Work Package
// — STAGE 2.1: Select IB via RL Policy — (Eqs. (16)-(17))
// Calculate selection probabilities P_bt for each IB based on historical performance (Eq. (16))
P_bt_all_IBs ← CalculateAllSelectionProbs (PerformanceMetrics_all_IBs(t-1))
IB_selected ← RouletteWheelSelection (IntelligenceBoxes, P_bt_all_IBs) // Select IB
// — STAGE 2.2: Improve WP_i using selected IB — (Eqs. (20-27))
X_new_ijt = X_ijt // Initialize candidate
switch IB_selected: // (Eqs. (20)-(27))
case CA: X_new_ijt ← U_r1_jt−1 + R_ijt
case WCA: X_new_ijt ← U_ijt−1 + R_ijt × U_r1_jt−1
…
case Restart: X_new_ijt ← Random (0, 1)
end switch
```

(continued on next page)

(*continued*)

```
// Decode X_new_ijt to Y_new_ijt, then evaluate FW_new, OC_new, and fitness
  Z_new_it* (Eqs. (11)-(15))
Y_new_ijt ← Decode(X_new_ijt)
FW_new, OC_new, Z_new_it* ← Evaluate(Y_new_ijt)
// — STAGE 2.3: Evaluate and Update Best & Current WP — (Eqs. (28)-(32))
if Z_new_it* ≥ Z_best(t−1): Z_best(t) ← Z_new_it*; U_best ← X_new_i // Update global
  best
else: Z_best(t) ← Z_best(t-1)
end if
// Check objective-wise improvement for accepting X_new_ijt for current WP_i
ω_it_1 ← (FW_new ≥ FW_i_current) ? 1 : 0 // FW improved/same (Eq. (29))
ω_it_2 ← (OC_new ≤ OC_i_current) ? 1 : 0 // OC improved/same (Eq. (30))
if (ω_it_1 + ω_it_2) ≥ 1: X_i ← X_new_i // Accept new solution (Eqs. (31)-(32))
// Update current objective values for WP_i
// else: Revert (X_i remains unchanged)
end if
end for // End WP_i loop
// — STAGE 2.4: Update RL Policy Rewards — (Eqs. (18)-(19))
for b = 1 to B: // For each IB
ImmediateReward_b ← CalculateImmediateReward (IB_b,
  PopulationChanges_this_iteration)
G_bt[b] ← Update_RL_Value(G_b(t-1), ImmediateReward_b, γ) // Update cumulative
  reward (Eq. (18))
UpdatePerformanceMetrics (IB_b, ImmediateReward_b, ImprovementValue) //
  Update Nb, Ab, Ib (Eq. (17) for Ib)
end for
end for // End main iteration loop
// — STAGE 3: Multi-objective Decision Making —
ParetoFrontSet ← FindNonDominatedSet (FinalPopulation) // Or all found non-
  dominated
BestSolution ← SelectByTOPSIS (ParetoFrontSet, ObjectivesWeights)
return ParetoFrontSet, BestSolution
End
```

4. Computational result

Our approach, as well as the methods we're comparing it with, were executed on a PC configured with an Intel Core i3 CPU at 3.70 GHz and 8 GB of DDR4 RAM, using Dev C++ for implementation. The computational outcomes presented in this study aim to highlight the contributions of our research. Initially, we'll carry out experiments based on the Taguchi design strategy, employing Design Expert software for this purpose. Following that, we propose to develop regression models for each method. These models will then be refined and compared across various techniques. We plan to discuss the experimental outcomes in a step-by-step manner, ensuring clarity and thorough understanding of the research's impact.

4.1. *Optimizing climbing perch aquaculture: a Taguchi design analysis of growth parameters and economic outcomes*

In this section, we delve into the experimental results obtained from the Taguchi design analysis, focusing on the relationship between various climbing perch growing parameters and their impact on two critical metrics: Fresh Weight (kg) and Operating Cost (Baht). Tables 8 and 9 compile and offer a detailed examination of how different growing conditions affect these metrics, laying the groundwork for enhancing climbing perch cultivation efficiency.

In this study, we explore the optimization of aquaculture practices, aiming to maximize yields while reducing operational costs. Documented in Table 8, we present various aquaculture treatments along with their respective outcomes, providing the essential data for our analysis.

Table 9, on the other hand, presents the estimated coefficients (from β_0 to β_{11}) for the regression models that predict Fresh Weight and Operating Cost based on the aforementioned growing parameters. These coefficients represent the expected change in Fresh Weight and Operating Costs for each unit change in a cultivation parameter, offering a statistical framework to optimize growing conditions for better yield and cost-efficiency.

The findings from our detailed examination of aquaculture conditions across 27 treatment runs are concisely summarized in Tables 10 and 11. Table 10 provides a snapshot of the range of environmental and management variables monitored during the study and their impact on the fresh weight of the fish, highlighting the upper and lower bounds of the observed data. In Table 11, the financial outcomes related to these parameters are presented, detailing the minimum and maximum costs

Table 8
Optimization of aquaculture parameters using Taguchi experimental design.

T	CPS (Species)	WT (°C)	DOL (Mg/L)	pH Levels	FS (Times/Day)	WQA (Mg/L)	WQNi (Mg/L)	WQNa (Mg/L)	LE (Hours/Day)	SD (Fish/L)	GP (Months)	Fresh Weight (Kg)	Operating Cost (Baht)
1	Testudineus	24	3	6	2	0.01	0.25	10	6	0.5	5	52.27	2590.45
2	Testudineus	24	3	6	3	0.10	0.50	20	9	1	7	53.09	3285.43
3	Testudineus	24	3	6	4	0.20	1.00	40	12	1.5	9	47.04	4038.18
4	Testudineus	27	5	7	2	0.01	0.25	20	9	1	9	48.15	3397.19
5	Testudineus	27	5	7	3	0.10	0.50	40	12	1.5	5	39.34	3420.85
6	Testudineus	27	5	7	4	0.20	1.00	10	6	0.5	7	61.66	2965.37
7	Testudineus	30	7	8	2	0.01	0.25	40	12	1.5	7	34.40	3532.61
8	Testudineus	30	7	8	3	0.10	0.50	10	6	0.5	9	54.60	3016.04
9	Testudineus	30	7	8	4	0.20	1.00	20	9	1	5	56.90	3104.12
10	Oligolepis	24	5	8	2	0.10	1.00	10	9	1.5	5	59.81	3046.14
11	Oligolepis	24	5	8	3	0.20	0.25	20	12	0.5	7	49.74	3257.12
12	Oligolepis	24	5	8	4	0.01	0.50	40	6	1	9	24.59	3036.05
13	Oligolepis	27	7	6	2	0.10	1.00	20	12	0.5	9	67.78	4082.10
14	Oligolepis	27	7	6	3	0.20	0.25	40	6	1	5	37.58	2844.64
15	Oligolepis	27	7	6	4	0.01	0.50	10	9	1.5	7	70.92	3641.22
16	Oligolepis	30	3	7	2	0.10	1.00	40	6	1	7	28.49	2885.08
17	Oligolepis	30	3	7	3	0.20	0.25	10	9	1.5	9	57.42	3333.27
18	Oligolepis	30	3	7	4	0.01	0.50	20	12	0.5	5	51.12	3224.64
19	Senegal	24	7	7	2	0.20	0.50	10	12	1	5	69.60	3406.68
20	Senegal	24	7	7	3	0.01	1.00	20	6	1.5	7	55.43	3253.18
21	Senegal	24	7	7	4	0.10	0.25	40	9	0.5	9	36.25	3463.99
22	Senegal	27	3	8	2	0.20	0.50	20	6	1.5	9	39.94	2880.99
23	Senegal	27	3	8	3	0.01	1.00	40	9	0.5	5	26.64	2833.27
24	Senegal	27	3	8	4	0.10	0.25	10	12	1	7	55.45	3284.61
25	Senegal	30	5	6	2	0.20	0.50	40	9	0.5	7	38.27	3245.62
26	Senegal	30	5	6	3	0.01	1.00	10	12	1	9	73.61	4106.44
27	Senegal	30	5	6	4	0.10	0.25	20	6	1.5	5	52.27	2875.46

Table 9
Hypothetical coefficients for predicting fresh weight of aquaculture species.

Coefficient	Fresh Weight (kg) Coefficient	Cost Coefficient
β_0 (Intercept)	71.287	52.460
β_1 (Species)	−0.148	−0.083
β_2 (Water Temperature)	−0.014	−0.027
β_3 (Dissolved Oxygen Levels)	2.000	1.493
β_4 (pH Levels)	−5.042	−4.082
β_5 (Feed Schedule)	0.972	0.851
β_6 (Water Quality Ammonia)	12.292	−8.520
β_7 (Water Quality Nitrites)	7.976	6.945
β_8 (Water Quality Nitrates)	−0.899	−0.009
β_9 (Light Exposure)	1.505	3.006
β_{10} (Stocking Density)	2.028	4.034
β_{11} (Growing Periods)	0.107	3.009

Note; T=Treatment, CPS=Climbing Perch Species, WT=Water Temperature, DSL=Dissolved Oxygen Levels, FS= Feeding Schedule, WQA=Water Quality-Ammonia, WQNi=Water Quality-Nitrites, WQNa=Water Quality-Nitrates, LE=Light Exposure, SD=Stocking Density, GP=Growing Periods, FW=Fresh Weight, OC=Operating Cost

associated with the aquaculture processes. These tables collectively offer valuable insights into the optimization of fish growth and cost management in aquaculture, serving as a practical reference for improved production strategies.

The results displayed in Table 10 (which includes 8 for fresh weight and 9 for cost) illustrate the range of conditions under which aquaculture operations were conducted and their corresponding outcomes in terms of both biological yield and economic cost. Table 10 reveals that the optimal conditions for fish growth vary across species and environmental factors, with fresh weight reaching a maximum under specific controlled conditions. It highlights the influence of variables such as water temperature, oxygen levels, and feeding schedules on the growth performance of different fish species. Table 11, on the other hand, shows the cost implications of these aquaculture conditions. It indicates that certain parameters, when optimized, can lead to a more cost-effective operation, whereas deviations from these optimal conditions can result in higher production costs. Together, these tables provide a clear indication that careful management of aquaculture conditions is not only crucial for the health and growth of the fish but also for the economic viability of the operation.

4.2. Comparative result of various methods for multi objective decision model

Our study aims to refine climbing perch aquaculture farming practices by examining how different growth conditions affect the climbing perch's chemical makeup. Table 12 is a key feature of our research, providing an in-depth look at how various optimization strategies stack up against each other when it comes to enhancing phytochemical levels. We use two main performance indicators in our analysis: Hyper Volume (HV) and Average Pareto Points (ARP). These metrics help us determine the effectiveness of each strategy while working within a set computational period of one hour.

The evaluation of optimization methods for Climbing Perch aquaculture, as shown in Table 12, was conducted over a standardized 60-minute computational period. The performance of each algorithm was assessed using three metrics: Hypervolume (HV), Average Ratio of Pareto-optimal Solutions (ARP), and the number of non-dominated versus dominated solutions. These indicators collectively offer a holistic view of each method's efficiency in generating diverse and optimal solution sets for the aquaculture objectives of maximizing fresh weight and minimizing operational cost.

Among all evaluated algorithms, the proposed RL-AMIS model outperformed the others, achieving the highest HV (0.63), ARP (1.73), and the largest number of non-dominated solutions (38 out of 100). The AMIS method followed closely with an HV of 0.61, ARP of 1.65, and 35 non-dominated solutions, while the MSA algorithm also showed competitive results. Other methods, including GA, EO, WCA, and CA, demonstrated moderate effectiveness, with HV values ranging from 0.56 to 0.59 and non-dominated solutions between 31 and 33. The inclusion of dominated versus non-dominated solution counts enhances result interpretability, underscoring RL-AMIS's strong capacity to identify optimal trade-offs and support intelligent, cost-effective decision-making in smart aquaculture systems.

Fig. 3(a) to 3(g) showcases the two-dimensional Pareto Fronts derived from different optimization methods applied in our study: Cultural Algorithm (CA), Water Cycle Algorithm (WCA), Memetic Algorithm (MSA), Equilibrium Optimizer (EO), Genetic Algorithm (GA), Adaptive Memory-inspired Strategy (AMIS), and Reinforcement Learning-Adaptive Memory-inspired Strategy (RL-AMIS). These figures represent the outcome of collecting Pareto points over a computational period of 60 minutes for each method.

The Pareto Front, in the context of optimization, is a set of points that represent nondominated solutions, considering multiple objectives. A solution is said to be nondominated if no other solution is better in all objectives. This concept is crucial for interpreting the results obtained from various methods. By analyzing the Pareto Fronts, we can visually assess the trade-offs between different objectives, such as maximizing phytochemical levels while minimizing resource use in Climbing Perch aquaculture. Each point on the Pareto Front represents a potentially optimal set of conditions, with none being absolutely better than the others across all criteria. Thus, the Pareto Front provides a valuable tool for decision-making, allowing researchers and cultivators to select an optimization strategy that best aligns with their specific goals and constraints.

4.3. Strategic optimization of climbing perch: evaluating efficiency through TOPSIS

This section details an analytical framework for optimizing finfish aquaculture, specifically focusing on climbing perch, through the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). Central to this discussion is the application of TOPSIS for evaluating and selecting the most efficient optimization strategies from a range of Pareto points gathered from various methods, as described in Section 4.2. The approach involves strategically adjusting the weights assigned to five critical objectives: Fresh weight (Kg) and Operation Cost (Baht), fundamental to the operation of climbing perch aquaculture.

Table 10
Range of aquaculture parameters and fresh weight outcomes.

	Climbing Perch Species	Water Temperature (°C)	Dissolved Oxygen Levels	pH Levels	Feed Schedule (times/day)	Water Quality Ammonia (mg/L)	Water Quality Nitrites (mg/L)	Water Quality Nitrates (mg/L)	Light Exposure (hours/day)	Stocking Density (fish/L)	Growing Periods (months)	Value of Fresh Weight (kg)
Min	Anabas Oligolepis	24	5	8	4	0.01	0.50	40	6	1	9	24.59
Max	Anabas Senegalensis	30	5	6	3	0.01	1.00	10	12	1	9	73.61

Table 11
Range of aquaculture parameters and associated costs.

	Climbing Perch Species	Water Temperature (°C)	Dissolved Oxygen Levels	pH Levels	Feed Schedule (times/day)	Water Quality Ammonia (mg/L)	Water Quality Nitrites (mg/L)	Water Quality Nitrates (mg/L)	Light Exposure (hours/day)	Stocking Density (fish/L)	Growing Periods (months)	Cost (Baht)
Min	Anabas Testudineus	24	3	6	2	0.01	0.25	10	6	0.5	5	2590.45
Max	Anabas Senegalensis	30	5	6	3	0.01	1.00	10	12	1	9	4106.44

Table 12
Comparative analysis of optimization methods for climbing perch aquaculture: Performance metrics and references.

Methods	CA	WCA	MSA	EO	GA	AMIS	RL-AMIS
References	[53]	[48]	[49]	[49]	[51]	[52]	Proposed model
HV	0.57	0.59	0.56	0.59	0.59	0.61	0.63
ARP	1.60	1.64	1.66	1.62	1.63	1.65	1.73
Non-Dominated Solutions	31	33	34	32	32	35	38
Dominated Solutions	69	67	66	68	68	65	62

Tables 13-15 serve as essential instruments in this comparative assessment, offering insights into how varying the emphasis between two core objectives—fish biomass and operational expenditure—can influence optimization results. These tables apply the TOPSIS method across three differentiated weighting scenarios: one with equal emphasis (0.5/0.5), one favoring maximum biomass output (0.9 for fresh weight), and another favoring economic restraint (0.9 for cost). The selection of these weights was strategic, intended to simulate decision-making patterns commonly encountered in aquaculture, where management objectives may fluctuate between maximizing growth, reducing expenses, or achieving a pragmatic compromise [54,55].

Rather than being randomly chosen, these weighting parameters were purposefully aligned with realistic stakeholder preferences, frequently observed in multi-criteria decision-making environments. In real-world settings, for instance, some aquaculture operators may prioritize maximizing yield to meet market demand, while others, operating under tighter resource constraints, may focus on minimizing production costs. Considering this spectrum of practical priorities strengthens the relevance and applicability of the study's findings to diverse production systems [56,57].

By evaluating the optimization results from multiple viewpoints, this work illustrates the flexibility of various algorithms—including the proposed RL-AMIS in handling contrasting operational priorities. Such an approach not only underlines the adaptability of the developed framework but also provides a practical tool for aquaculture planning. It supports more precise alignment between strategic goals and operational decisions, thereby making the trade-off dynamics more explicit and reinforcing the model's value as a credible, decision-oriented support system [58].

In the balanced approach, RL-AMIS emerges as the leading method, delivering the highest fresh weight (71.44 kg) with the lowest operating cost (2635.41 Baht), indicating its superior capability to optimize both objectives simultaneously. This is closely followed by AMIS, which also performs exceptionally but with slight compromises in both fresh weight and cost. The other methods, including CA, WCA, MSA, EO, and GA, show competitive but lesser efficiency, highlighting the importance of sophisticated optimization techniques in achieving a balanced outcome.

When the emphasis shifts towards maximizing fresh weight, the RL-AMIS method again tops the chart with an impressive fresh weight of 68.16 kg, albeit at a slightly higher cost (2454.41 Baht). This reflects RL-AMIS's adaptability to prioritize growth outcomes effectively. AMIS remains strong in this scenario, reinforcing its position as the second-best method. This table demonstrates that while focusing on fresh weight can lead to slight increases in costs, the advanced algorithms like RL-AMIS and AMIS still manage to keep these costs relatively low compared to others.

Focusing predominantly on cost efficiency, RL-AMIS proves to be the most cost-effective method, achieving the lowest operating cost (2718.44 Baht) while still ensuring a commendable fresh weight of 75.96 kg. This highlights RL-AMIS's exceptional ability to minimize costs without substantially sacrificing growth performance. AMIS follows, aligning with the trend observed in previous tables, showcasing its effectiveness but with a slight reduction in optimization performance compared to RL-AMIS.

Across all tables, RL-AMIS stands out as the most effective method for optimizing climbing perch aquaculture, capable of achieving superior outcomes regardless of the optimization focus. AMIS secures a solid second place, consistently outperforming the other methods but not quite matching RL-AMIS's efficacy. The rest of the methods, while effective to varying degrees, exhibit more significant trade-offs between fresh weight and cost, underscoring the advanced capabilities of RL-AMIS and AMIS in navigating these trade-offs.

These analyses underscore the nuanced understanding required in aquaculture optimization, where strategic adjustments in optimization priorities can lead to different but equally valuable outcomes. RL-AMIS, with its top-tier performance across scenarios, offers a versatile and powerful tool for cultivators aiming to balance growth and cost efficiency in climbing perch farming.

4.4. Validation of optimized aquaculture parameters for climbing perch: a comparative study of theoretical predictions and actual outcomes

This study explores the practical application of optimized parameters for climbing perch aquaculture, aiming to validate theoretical models against actual fish production outcomes. Focusing on fresh weight and operating costs with equal importance, we conducted seven growing cycles under identical, optimized conditions derived from prior research. The objective was to assess the accuracy of theoretical predictions in real-world settings, examining the potential for achieving or surpassing expected yields and cost-efficiencies. This analysis provides valuable insights into the effectiveness of optimization strategies in climbing perch cultivation, contributing to the ongoing refinement of aquaculture practices. The result is shown in Table 16.

The results presented in Table 16 provide a comparative analysis between theoretical expectations and empirical outcomes in climbing perch aquaculture. Both fresh weight and operating cost were evaluated with equal importance across seven consecutive growing cycles conducted under identical optimized conditions identified through prior experiments. This approach yields a robust assessment of the practical applicability and reproducibility of the optimized parameters.

The theoretical benchmarks predicted a fresh weight of 71.44 kg and an operating cost of 2635.41 Baht. Experimental results showed an average fresh weight of 67.04 kg and an average operating cost of 2888.16 Baht. The percent differences 6.16 % for fresh weight and 9.59 % for operating cost reflect a close alignment between predicted and observed outcomes. Standard deviations were relatively low (2.02 kg for weight and 76.84 Baht for cost), indicating consistent performance

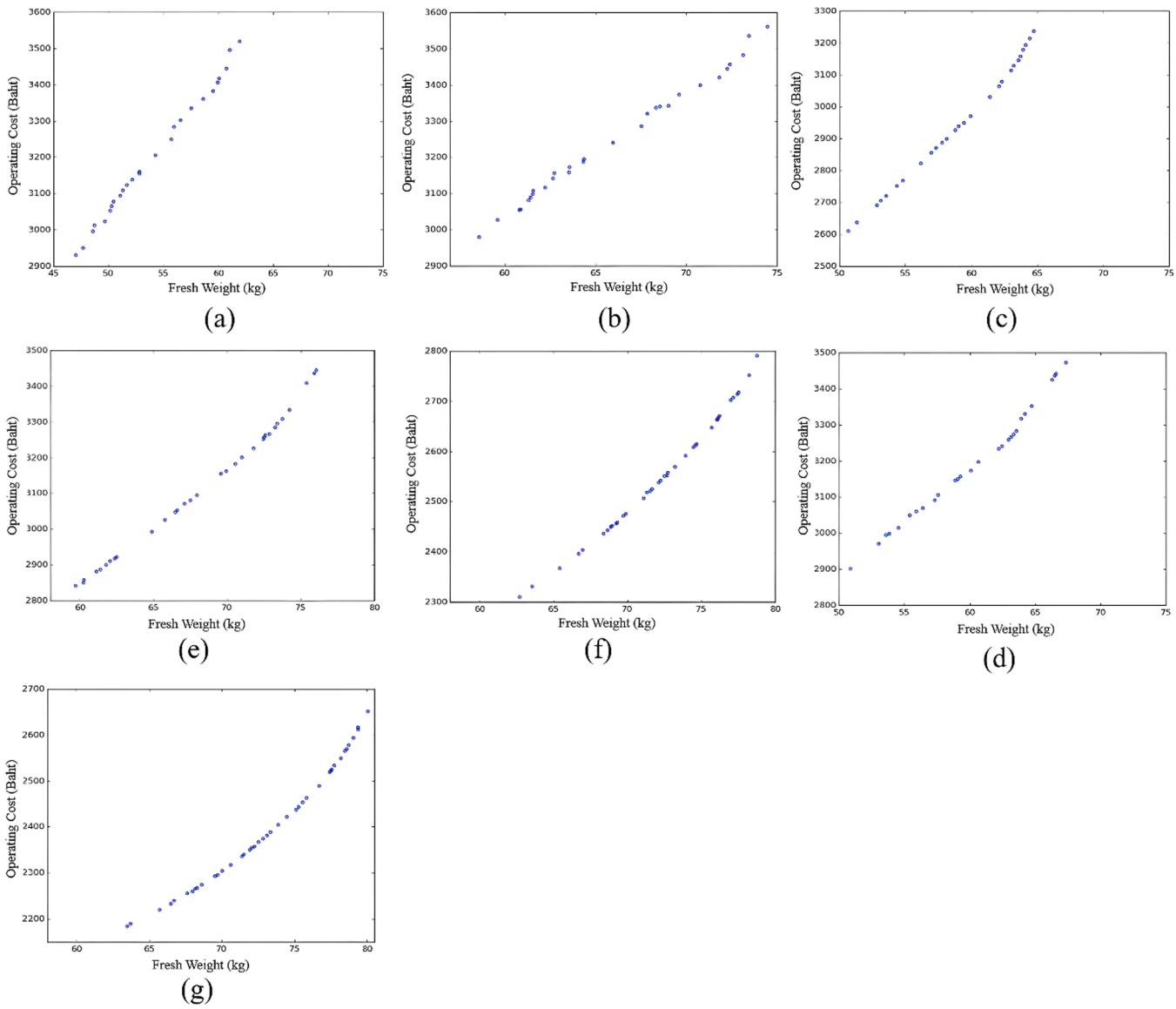

Fig. 3. 2D-Pareto front of different optimization methods.

Table 13
Balanced weighting approach (Fresh weight 0.5, Cost 0.5).

Value/Method	CA	WCA	MSA	EO	GA	AMIS	RL-AMIS
Fresh weight (kg)	55.37	62.79	54.95	57.43	63.74	68.78	71.44
Operating Cost (Baht)	3522.91	3461.33	3177.53	3364.34	3103.46	2710.70	2635.41
Climbing Perch Species	A. Testudineus	A. Oligolepis	A. Senegalensis	A. Testudineus	A. Senegalensis	A. Oligolepis	A. Testudineus
Water Temperature (°C)	27	27	27	30	24	24	24
Dissolved Oxygen Levels	5	7	3	7	7	5	3
pH Levels	7	6	8	8	7	8	6
Feed Schedule (times/day)	2	2	2	3	3	2	2
Water Quality Ammonia (mg/L)	0.01	0.1	0.2	0.1	0.01	0.1	0.1
Water Quality Nitrites (mg/L)	0.25	1	0.5	0.5	1	1	0.25
Water Quality Nitrates (mg/L)	20	20	20	10	20	10	10
Light Exposure (hours/day)	9	12	6	6	6	9	6
Stocking Density (fish/L)	1	0.5	1.5	0.5	1.5	1.5	0.5
Growing Periods (months)	9	9	9	9	7	5	5

across all replications.

Fresh weight ranged from 63.03 kg to 69.01 kg, with Crop 5 demonstrating the highest yield, suggesting that in certain cycles, real-world production can closely approach or even exceed theoretical predictions. This supports the effectiveness of the optimized environmental and operational settings.

In terms of cost, the observed values ranged from 2825.21 Baht to 3053.61 Baht. Although slightly higher than expected, these variations

Table 14
Fresh weight emphasis (Fresh weight 0.1, Cost 0.9).

Value/Method	CA	WCA	MSA	EO	GA	AMIS	RL-AMIS
Fresh weight (kg)	51.92	60.29	51.68	57.04	63.00	67.63	68.16
Operating Cost (Baht)	3243.77	3142.24	2732.80	3101.36	2954.33	2584.73	2454.41
Climbing Perch Species	A. Testudineus	A. Oligolepis	A. Senegalensis	A. Testudineus	A. Senegalensis	A. Oligolepis	A. Testudineus
Water Temperature (°C)	27	27	27	30	24	24	24
Dissolved Oxygen Levels	5	7	3	7	7	5	3
pH Levels	7	6	8	8	7	8	6
Feed Schedule (times/day)	2	2	2	3	3	2	2
Water Quality Ammonia (mg/L)	0.01	0.1	0.2	0.1	0.01	0.1	0.1
Water Quality Nitrites (mg/L)	0.25	1	0.5	0.5	1	1	0.25
Water Quality Nitrates (mg/L)	20	20	20	10	20	10	10
Light Exposure (hours/day)	9	12	6	6	6	9	6
Stocking Density (fish/L)	1	0.5	1.5	0.5	1.5	1.5	0.5
Growing Periods (months)	9	9	9	9	7	5	5

Table 15
Cost efficiency emphasis (Fresh weight 0.9, Cost 0.1).

Value/Method	CA	WCA	MSA	EO	GA	AMIS	RL-AMIS
Fresh weight (kg)	58.98	67.72	59.93	64.32	70.54	72.97	75.96
Operating Cost (Baht)	3604.47	3665.74	3146.30	3404.86	3147.83	3888.23	2718.44
Climbing Perch Species	A. Testudineus	A. Oligolepis	A. Senegalensis	A. Testudineus	A. Senegalensis	A. Oligolepis	A. Testudineus
Water Temperature (°C)	27	27	27	30	24	24	24
Dissolved Oxygen Levels	5	7	3	7	7	5	3
pH Levels	7	6	8	8	7	8	6
Feed Schedule (times/day)	2	2	2	3	3	2	2
Water Quality Ammonia (mg/L)	0.01	0.1	0.2	0.1	0.01	0.1	0.1
Water Quality Nitrites (mg/L)	0.25	1	0.5	0.5	1	1	0.25
Water Quality Nitrates (mg/L)	20	20	20	10	20	10	10
Light Exposure (hours/day)	9	12	6	6	6	9	6
Stocking Density (fish/L)	1	0.5	1.5	0.5	1.5	1.5	0.5
Growing Periods (months)	9	9	9	9	7	5	5

Table 16
Comparative analysis of theoretical and empirical phytochemical concentrations in RL-AMIS-derived climbing perch fishery.

	Fresh weight (kg)	Operating cost (฿)
Theory	71.44	2635.41
Crop 1	66.87	2825.21
Crop 2	67.30	2863.43
Crop 3	68.37	2854.61
Crop 4	66.21	2863.66
Crop 5	69.01	3053.61
Crop 6	68.46	2906.21
Crop 7	63.03	2850.37
Average	67.04	2888.16
Standard Deviation	2.02	76.84
Percent Different from Theory	6.16	9.59

are reasonable considering potential real-world influences such as equipment calibration drift, localized microclimate fluctuations, or unmodeled biological stressors (e.g., feed competition or behavioral changes).

Overall, the minimal variance from theoretical models confirms the robustness of the RL-AMIS optimization framework. This strong correlation between predicted and actual outcomes validates its use as a practical tool for aquaculture planning and operation. The findings also suggest that with further refinement and adaptive control, even greater precision and efficiency in climbing perch cultivation may be achievable across various farming scales and environmental contexts.

5. Discussion

5.1. Advancements in optimization methods: the superiority of RL-AMIS

The RL-AMIS method emerges as a groundbreaking advancement in the sphere of optimization techniques for Climbing Perch Aquaculture. Its exemplary performance, as evidenced by leading scores in Hyper Volume (HV) and Average Pareto Points (ARP), attests to its unparalleled efficiency. These metrics are indispensable in gauging an optimization strategy's prowess in balancing multifaceted objectives effectively, demonstrating RL-AMIS's proficiency in producing a wide array of optimal solutions. This capacity highlights its versatility and precision in navigating the complexities inherent in aquaculture systems, suggesting a substantial potential for transforming Climbing Perch Aquaculture through sophisticated, multi-objective optimization [52].

RL-AMIS's supremacy is further elucidated through a comparative analysis with other methodologies such as AMIS, MSA, GA, EO, WCA, and CA. It is RL-AMIS's integrated approach, combining reinforcement learning with artificial intelligence, that bestows it with a dynamic adaptability to the unique challenges of aquaculture. This hybrid model significantly outperforms conventional methods by offering tailored strategies that align with the specific environmental and operational demands of Climbing Perch Aquaculture. Such a nuanced approach ensures the method's responsiveness to a broad spectrum of aquacultural conditions, from water quality variables to species-specific nutritional needs, thereby endorsing its robustness in optimizing operational efficiency and production efficacy [49].

The findings of this study have direct practical implications. For farmers, the RL-AMIS framework offers a decision-support tool that identifies cost-efficient environmental and operational conditions, enabling data-driven planning and minimizing trial-and-error practices. For policymakers, the model provides quantitative insights to guide the

development of evidence-based guidelines or subsidies that promote efficient and sustainable fish farming. Additionally, system designers can leverage the modular optimization logic to develop smart aquaculture platforms that autonomously respond to fluctuating farm conditions, enhancing system resilience and scalability.

The RL-AMIS framework developed in this study demonstrates promising potential for scalability and generalization across aquaculture systems. Due to its modular design and adaptability to various input parameters, RL-AMIS can be calibrated for different species by updating the experimental design and training it with species-specific growth, environmental, and cost data. Furthermore, its decision-fusion mechanism is applicable in larger-scale production settings where real-time optimization is required across multiple ponds or sites. This makes RL-AMIS a versatile tool not only for climbing perch but also for sustainable and cost-efficient farming of species such as tilapia, catfish, or shrimp, especially in precision aquaculture environments.

5.2. Operational and environmental considerations in aquaculture optimization

The influence of environmental and management variables on Climbing Perch Aquaculture cannot be overstated. From water quality and pH levels to feed schedules and stocking densities, each factor plays a crucial role in determining the success of cultivation efforts. The detailed examination of how these conditions affect fresh weight and operating cost, as presented in Tables 8 and 9, aligns with broader research emphasizing the critical nature of optimizing these parameters for sustainable aquaculture practices. Understanding the interplay of these variables is key to advancing cultivation techniques that are both productive and environmentally responsible [3,40].

The pursuit of optimization in Climbing Perch Aquaculture necessitates a multi-objective approach, balancing the dual goals of maximizing yield and minimizing costs. The application of techniques such as Pareto Front Optimization (PFO) and TOPSIS is instrumental in navigating the complexities of this trade-off, offering a framework for decision-making that considers the full spectrum of operational objectives. This emphasis on multi-objective optimization underscores the sector's move towards more nuanced and sophisticated methods of management, reflecting a growing acknowledgment of the need for strategies that can adapt to the multifarious demands of aquaculture operations [19,37].

Through these subtopics, the discussion provides an in-depth analysis of the current state of optimization in Climbing Perch Aquaculture, highlighting both the technological advancements and the critical operational considerations that underpin successful cultivation practices. This nuanced exploration not only contributes to a deeper understanding of the sector's challenges and opportunities but also charts a course for future research and implementation in the field.

5.3. Implications for aquaculture efficiency and sustainability

This study's exploration into the use of the RL-AMIS method for Climbing Perch Aquaculture optimization carries significant implications, not only for the aquaculture industry but also for broader applications in resource management and sustainability practices. The demonstrated superiority of RL-AMIS in achieving high performance across Hyper Volume (HV) and Average Pareto Points (ARP) metrics offers a glimpse into the future of aquaculture operations, where advanced computational methods could lead to enhanced efficiency, productivity, and environmental sustainability.

The application of RL-AMIS has shown that it is possible to balance growth and economic efficiency in a manner that could revolutionize Climbing Perch Aquaculture, setting a benchmark for operational excellence within the sector. This method's success in generating diverse and optimal solutions underlines its potential in contributing to the sustainability of aquaculture practices by optimizing resource use and minimizing environmental impact. The adaptability and precision of RL-AMIS also suggest that this approach could be customized for various species and environmental conditions, indicating a wide range of applicability across different aquaculture systems.

Moreover, the principles underlying the RL-AMIS method could be generalized beyond Climbing Perch Aquaculture to encompass a broader array of aquaculture endeavors. Its flexibility and effectiveness in addressing multiple objectives simultaneously make it a valuable tool for enhancing productivity and sustainability across the sector. Additionally, the optimization strategies employed by RL-AMIS hold potential for cross-disciplinary applications, including agriculture, environmental management, and industrial processes, where managing complex trade-offs is a common challenge.

This study not only sheds light on the efficacy of the RL-AMIS method in optimizing Climbing Perch Aquaculture but also opens up new avenues for applying similar methodologies across various fields to address pressing sustainability and efficiency challenges. The ongoing development and application of these advanced optimization techniques are essential for driving forward innovation in aquaculture and beyond, ensuring the sustainable management of global resources.

Although the study demonstrates promising results under real-world smart aquaculture conditions, it does not include a traditional baseline system operated without AI-based support. This decision was made to focus on evaluating relative algorithmic performance within consistent farming contexts. Future investigations may benefit from incorporating such control setups to provide a broader frame of reference, especially in quantifying the added value of intelligent optimization under diverse production systems. Doing so would further strengthen the evidence base for applying decision-support tools like RL-AMIS in both conventional and technology-enhanced aquaculture.

5.4. Contribution to Environmental Sustainability

This study provides a concrete response to the ongoing global challenge of achieving both economic efficiency and environmental responsibility in aquaculture. The RL-AMIS system, while designed to reduce operational expenditures, also plays a meaningful role in promoting environmental sustainability through more thoughtful resource use. For instance, the model identifies farming conditions that help limit overfeeding and nitrogen-based waste, thus improving water quality and reducing the need for frequent replacement or treatment an issue that has been widely discussed in the literature, including works by Tisdell [33] and Bohnes et al. [34]. Moreover, by dynamically adjusting parameters such as light exposure and oxygen levels based on real-time optimization, the system conserves energy and contributes to lowering greenhouse gas emissions. These improvements align with the direction proposed by Carballeira Braña et al. [31] and Sampantamit et al. [36], who emphasized the importance of localized and adaptive management practices. In this light, RL-AMIS offers not only a technological enhancement but also a practical strategy for balancing productivity with ecological protection—an approach that closely reflects the sustinability model outlined by Diana et al. [37].

6. Conclusion

In addressing the urgent need for enhanced Climbing Perch Aquaculture productivity and cost-efficiency, this research ventured into uncharted territories to bridge a crucial gap in aquaculture practices. The primary problem statement revolved around identifying and deploying advanced optimization strategies capable of harmonizing the dual objectives of maximizing growth while ensuring economic and environmental sustainability. This investigation was propelled by the observed research gap in effectively integrating a multitude of complex growth determinants towards the holistic optimization of cultivation practices.

This research advanced Climbing Perch Aquaculture optimization by deploying a novel methodology that combines Taguchi experimental

design and the RL-AMIS method. Initially, Taguchi design, executed through Design Expert software, structured experiments to identify key factors affecting aquaculture efficiency. This step laid the groundwork for developing a multi-objective regression model that quantifies the impact of various parameters on growth and costs. Subsequently, the RL-AMIS method— an enhancement of the traditional AMIS by integrating reinforcement learning—was employed for optimization. This integration allows for an adaptive and intelligent exploration of solutions, leveraging techniques like TOPSIS and Pareto Front Optimization to simultaneously address productivity and cost-efficiency.

The core of our findings underscored the unparalleled efficacy of the RL-AMIS method, which achieved remarkable Hyper Volume (HV) scores of 0.63 and Average Pareto Points (ARP) of 1.73, decisively outperforming competing optimization methods such as AMIS, CA, WCA, MSA, EO, and GA. These numerical results not only validated the superior capability of RL-AMIS in navigating the intricate optimization landscape but also highlighted its potential to revolutionize Climbing Perch Aquaculture by generating diverse, optimal solutions for a range of cultivation challenges.

The implications of this research are profound, extending beyond academic interest to practical applications within the aquaculture industry. By demonstrating the effectiveness of RL-AMIS in refining Climbing Perch cultivation towards greater efficiency and sustainability, the study contributes valuable insights toward the global endeavor of sustainable aquaculture development. It underscores the pivotal role of cutting-edge computational techniques in resolving the complex interplay between aquaculture productivity and ecological stewardship, suggesting a model that can be adapted across various aquaculture systems for broader impact.

Future research directions are abundant and promising, inviting exploration into the scalability of RL-AMIS across different species, environments, and operational scales. Further inquiries might delve into the impacts of varying environmental variables on aquaculture yields, the integration of additional optimization metrics, and the practical applicability of these methodologies in real-world settings. A comparative analysis with traditional cultivation methods could further elucidate the practical advantages and limitations of advanced computational optimization techniques, offering a comprehensive roadmap for their implementation in enhancing the global aquaculture landscape.

Ethical statement

Ethical approval

The research project, code 01-67-006, has been approved by the Institutional Animal Care and Use Committee at Rajamangala University of Technology Isan. The protocol complies with the Ethics of Use of Animals for Scientific Work by the National Research Council of Thailand. The study will be conducted at the Department of Plant Science, Faculty of Natural Resources, Sakon Nakhon Campus. The approval ensures adherence to national ethical standards and is signed by the University President. A copy of the ethical approval certificate is attached as supplementary material for verification.

Informed Consent

Not applicable. This study does not involve human participants.

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General ethical compliance

All procedures involving animals were conducted in accordance with the Ethical Principles and Guidelines for the Use of Animals for Scientific Purposes set forth by the National Research Council of Thailand. The study was carried out under strict compliance with institutional guidelines for animal care and use.

CRediT authorship contribution statement

Kosit Sriputhorn: Writing – original draft, Data curation. **Achara Jutagate:** Writing – review & editing, Validation. **Surasak Matitopanum:** Writing – review & editing, Supervision, Project administration. **Rungwasun Kraiklang:** Visualization. **Rapeepan Pitakaso:** Writing – review & editing, Validation. **Chakat Chueadee:** Methodology. **Sarayut Gonwirat:** Software, Methodology.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper. The research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2025.101108.

Data availability

Data will be made available on request.

Reference

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